Identification of mutation hotspots to measure carcinogenic risk in Barrett's Esophagus using seq.hotSPOT and compSPOT

Sydney R. Grant[1,2]; Spencer R. Rosario[3]; Ella M. Sampson[1]; Barbara A. Foster[4]; Lei Wei[3]; Wendy J. Huss[1,2]; Gyorgy Paragh[1,2]
[1]Department of Dermatology [3]Department of Biostatistics & Bioinformatics
[2]Department of Cell Stress Biology [4]Department of Pharmacology & Therapeutics
Roswell Park Comprehensive Cancer Center, Buffalo, NY 14263

INTRODUCTION

- An estimated 21,560 new cases of esophageal cancer will be diagnosed in 2023 (SEER, 2023)
- Barrett's Esophagus (BE) is a precursor lesion to Esophageal Adenocarcinoma (ESAD), with a reported risk of progression ranging from 0.2-11.3% per patient per year (Gaddam et al., Hvid-Jensen et al.)
- Previous studies comparing progressive BE carries significantly more mutations in ESAD-associated genes, particularly in *TP53* (Paulson et al.)

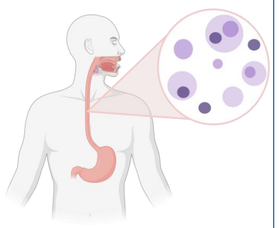

Figure 1. Clonal mutations in known cancer-causing genes present in clinically-normal esophagus

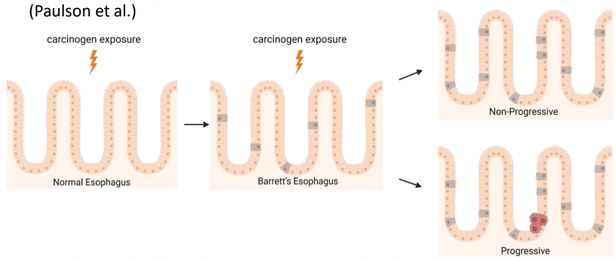

Figure 2. Schematic of healthy esophagus progression to Barrett's Esophagus followed by possible progression to Esophageal Adenocarcinoma

Rationale and Significance

- There is a lack of tools to accurately identify BE likely to progress to advanced disease.
- seq.hotSPOT and compSPOT can help identify relevant genomic regions for measuring carcinogenic progression in BE.
- Identifying robust markers of disease progression is essential for the development of reliable tests for assessing cancer risk.
- Previous work from our lab has utilized mutation hotspots found in bladder cancers to aid prediction of cancer risk in normal urothelium (Grant et al.).

MATERIALS & METHODS

Esophageal Cancer Datasets (cBioPortal):
Datasets containing DNA-sequencing information from tumor samples of ESAD and ESCC were downloaded from the cBioPortal database and curated based on Figure 3 parameters. Mutations were filtered to only contain those sequenced by MSK-IMPACT and Sanger Institute targeted sequencing panels.

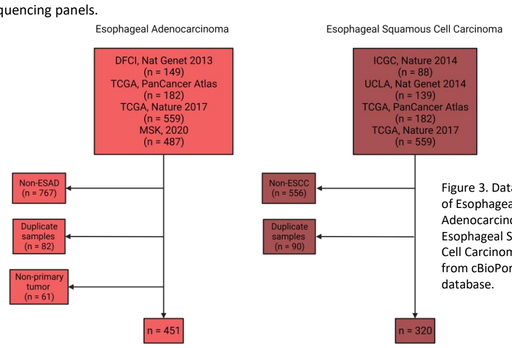

Figure 3. Dataset curation of Esophageal Adenocarcinoma and Esophageal Squamous Cell Carcinoma datasets from cBioPortal database.

Barrett's Esophagus Datasets (Paulson et al., Stachler et al.):
Barrett's Esophagus samples from 80 BE patients with progression (n = 40) or no progression (n = 40) to ESAD. Biopsies collected from patients at multiple timepoints with a median 17.5 year follow-up. Validation dataset from 86 BE patients with progression (n = 10) or no progression (n = 76) to ESAD or high-grade dysplasia. Biopsies taken at single time point.
Normal Esophagus Dataset (Martincorena et al.):
844 small sections of clinically-normal esophagus were sequenced using deep targeted sequencing from 9 deceased organ donors with no history of esophageal disease.
Hotspot Discovery:
The combined ESAD and ESCC datasets were separately input into the seq.hotSPOT Bioconductor package in R to identify the top 10% most mutated regions in each esophageal cancer subtype. Significant mutation hotspots were identified using compSPOT.
Statistical Analysis:
Pearson Correlation and Wilcoxon Rank Sum Test were used for statistical comparisons using stats package in R.
Neural Network Classification Model:
Model training, parameter tuning, and performance measurement were preformed using caret, neuralnet, and pROC packages. Relative variable importance was calculated using clusterGeneration package.
Images were generated using biorender.com. Figures created using R packages ggplot2, ggpubr, ggrepel, ggsignif, ggplotify, and cvms. All computational analyses were performed in R version 4.3.1.

seq.hotSPOT

Figure 4A. amp_pool function uses input mutation dataset and user defined amplicon length to search the genome for densely mutated areas. Mutations within one amplicon length of the neighboring mutation are clustered together.

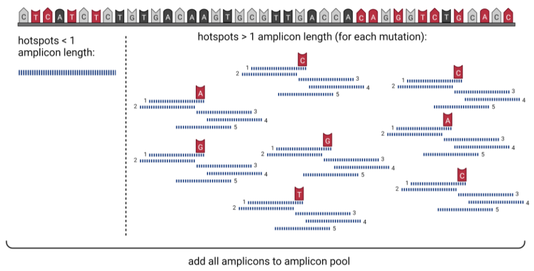

Figure 4B. For spots less than one amplicon length, amp_pool function creates a single weighted amplicon. For spots greater than one amplicon length, amplicons at five different positions around each mutation are created. All created amplicons are added to an amplicon pool.

Figure 4C. fw_hotspot function uses the mutation dataset, amplicon pool generated from amp_pool, amplicon length, and size of desired panel to find a subset of amplicons to cover each mutation spot. Spots less than one amplicon length are covered with single amplicon. For spots greater than one amplicon length, the amplicon that covers the greatest number of mutations is pulled from the pool, then continuously add the amplicon that covers the greatest number of additional mutations until the entire spot is covered.

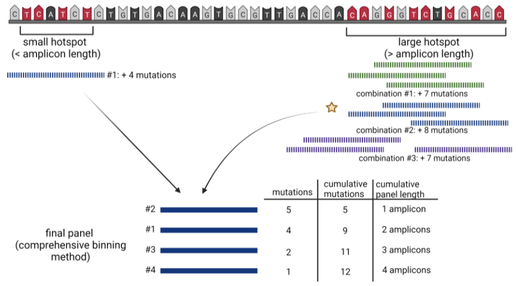

Figure 4D. com_hotspot function uses the mutation dataset, amplicon pool generated from amp_pool, panel developed by fw_hotspot, amplicon length, and size of desired panel to find a subset of amplicons to cover each mutation spot. Spots less than one amplicon length are covered with single amplicon. For spots greater than one amplicon length, every possible combination of amplicons to cover that spot is tested. Combination yielding the highest total mutation count is added to the final panel. All chosen amplicons in final panel are ranked based on mutation frequency.

compSPOT

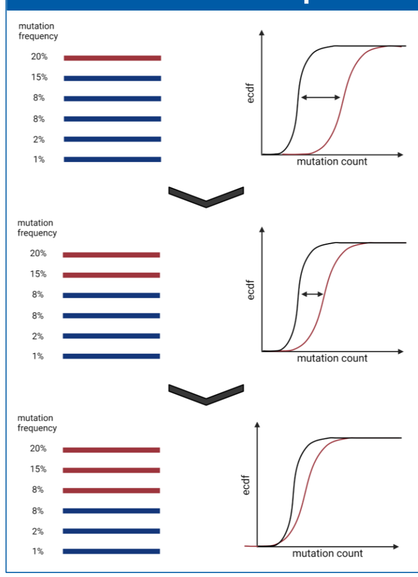

Figure 4E. sig.spots function from compSPOT package inputs mutation dataset, list of genomic regions created by seq.hotSPOT, and a threshold p-value and empirical distribution function distance.

Beginning with the top ranked genomic region, the mutation frequency per region is calculated for the top region and all lower regions.

The frequency per sample is compared between the top and bottom regions using a Kolmogorov-Smirnov test. If the p-value and distance calculated by the Kolmogorov-Smirnov test meets the threshold requirements set by the user, this region will be added to a list of significant hotspots.

The algorithm is then rerun, continuously adding an additional region until the threshold p-value or distance are no longer met.

RESULTS

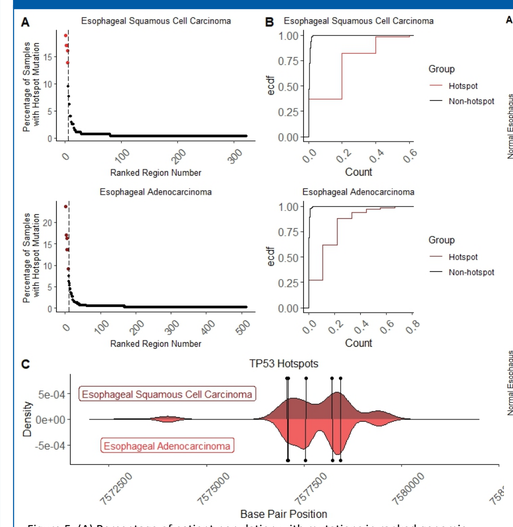

Figure 5. (A) Percentage of patient population with mutations in ranked genomic segments and statistically significant hotspots (colored). (B) Empirical Cumulative Distribution Function of hotspots and non-hotspots in ESAD and ESCC identified by compSPOT. (C) Mutation distribution on TP53 in ESAD and ESCC, lollipop plot indicates location of mutation hotspots. Data based on cBioPortal datasets.

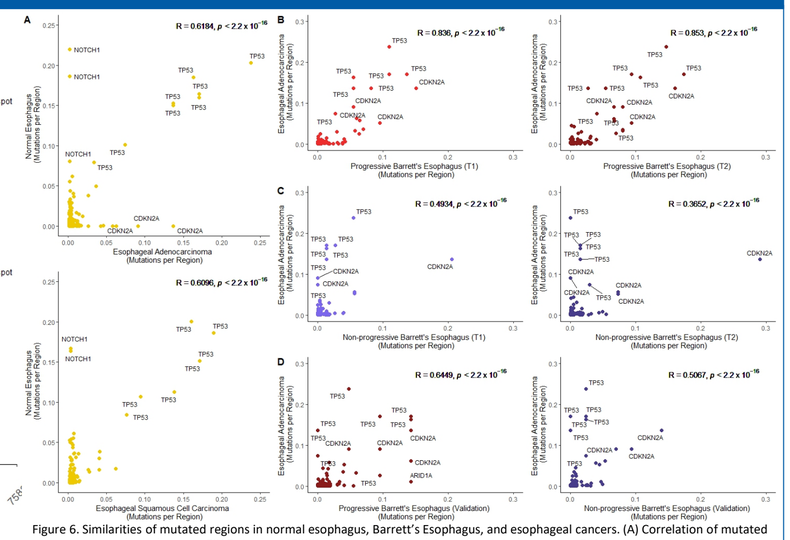

Figure 6. Similarities of mutated regions in normal esophagus, Barrett's Esophagus, and esophageal cancers. (A) Correlation of mutated regions in normal esophagus vs ESAD (top) and ESCC (bottom). (B) Correlation of mutated regions in ESAD vs progressive BE at initial biopsy (left) and biopsy at time of ESAD diagnosis (right). (C) Correlation of mutated regions in ESAD vs non-progressive BE at initial biopsy (left) and follow-up biopsy (right). (D) Correlation of ESAD vs validation cohort of progressive and non-progressive BE. Correlations calculated using Pearson Correlation. Data based on cBioportal, Martincorena et al., Paulson et al., and Stachler et al. datasets.

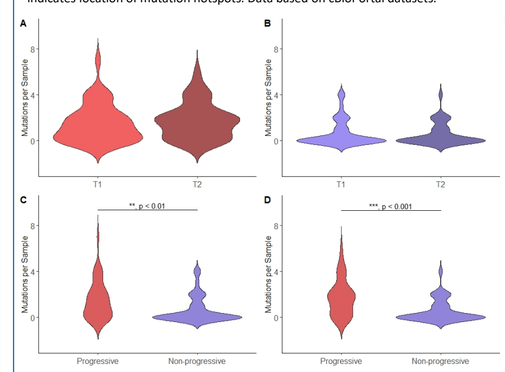

Figure 7. Comparison of mutations per sample in ESAD significant hotspots (A) Progressive BE timepoint 1 vs timepoint 2. (B) Non-progressive BE timepoint 1 vs timepoint 2. (C) Progressive BE vs non-progressive BE at timepoint 1, and (D) progressive vs non-progressive BE at timepoint 2. Significance calculated by Wilcoxon Rank Sum Test. Data based on Paulson et al., dataset.

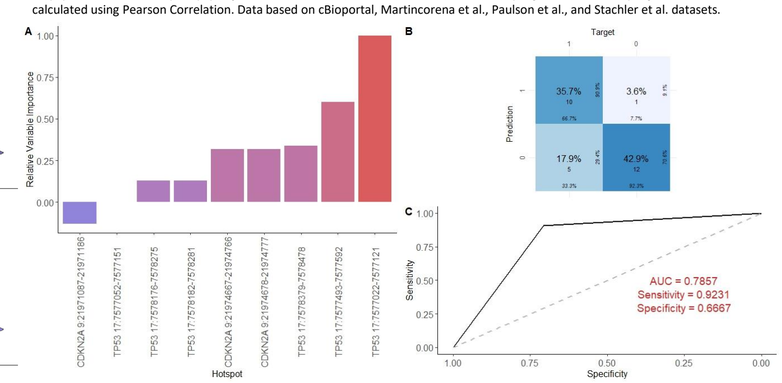

Figure 8. Neural network classification model for prediction of esophageal disease progression. Model performance parameters based on classification of test dataset. (A) Variable importance summary of 9 mutation hotspots in ESAD. (B) Confusion matrix of prediction accuracy on test dataset. (C) ROC curve of model performance on test data. Data based on Paulson et al. dataset.

CONCLUSIONS & FUTURE DIRECTIONS

- seq.hotSPOT and compSPOT provide an efficient pipeline for identifying genomic areas of high mutation sensitivity.
- There are large similarities in highly mutated genomic areas in Esophageal Adenocarcinoma, Esophageal Squamous Cell Carcinoma, and clinically normal esophagus.
- Mutation patterns in patients with progressive Barrett's Esophagus show higher correlation to Esophageal Adenocarcinoma compared to patients with no observed progression, even many years prior to cancer diagnosis.
- Mutation hotspots in Esophageal Adenocarcinoma identified with compSPOT show significantly higher mutation enrichment in progressive Barrett's Esophagus compared to non-progressive and may be used in a machine learning model to predict risk of esophageal disease progression.
- Future studies with larger cohort sizes may identify additional significant hotspot markers for the prediction of disease progression.
- Addition of personal risk factors of Esophageal Adenocarcinoma as features in machine learning model may increase accuracy of classification algorithm.
- Similar approaches may be taken for cancer risk assessment in additional epithelial organ sites.

REFERENCES

Cancer Stat Facts: Esophageal Cancer (2023), National Cancer Institute, Surveillance, Epidemioogy, and End Results Program
Gaddam S et al., Persistence of nondysplastic Barrett's esophagus identifies patients at lower risk for esophageal adenocarcinoma: results from a large multicenter cohort. Gastroenterology. 2013 Sep;145(3):548-53.e1. doi: 10.1053/j.gastro.2013.05.040. Epub 2013 May 25. PMID: 23714382.
Hvid-Jensen F et al., Incidence of adenocarcinoma among patients with Barrett's esophagus. N Engl J Med. 2011 Oct 13;365(15):1375-83. doi: 10.1056/NEJMoa1103042. PMID: 21995385.
Paulson, T. G. et al., (2022). Somatic whole genome dynamics of precancer in Barrett's esophagus reveals features associated with disease progression. Nature communications, 13(1), 2300. https://doi.org/10.1038/s41467-022-29767-7
Stachler, M. D. et al., (2018). Detection of Mutations in Barrett's Esophagus Before Progression to High-Grade Dysplasia or Adenocarcinoma. Gastroenterology, 155(1), 156–167. https://doi.org/10.1053/j.gastro.2018.03.047
Martincorena, I. et al., (2018). Somatic mutant clones colonize the human esophagus with age. Science (New York, N.Y.), 362(6417), 911–917. https://doi.org/10.1126/science.aau3879
Grant, S. R., et al., (2023). HotSPOT: A Computational Tool to Design Targeted Sequencing Panels to Assess Early Photocarcinogenesis. Cancers, 15(5), 1612. https://doi.org/10.3390/cancers15051612
Grant, S. R. et al., (2023). Mutation Hotspots Found in Bladder Cancer Aid Prediction of Carcinogenic Risk in Normal Urothelium. International journal of molecular sciences, 24(9), 7852. https://doi.org/10.3390/ijms24097852
Dulak, A. M. et al., (2013). Exome and whole-genome sequencing of esophageal adenocarcinoma identifies recurrent driver events and mutational complexity. Nature genetics, 45(5), 478–486. https://doi.org/10.1038/ng.2591
Cancer Genome Atlas Research Network, Analysis Working Group: Asan University, BC Cancer Agency, Brigham and Women's Hospital, Broad Institute, Brown University, Case Western Reserve University, Dana-Farber Cancer Institute, Duke University, Greater Poland Cancer Centre, Harvard Medical School, Institute for Systems Biology, KU Leuven, Mayo Clinic, Memorial Sloan Kettering Cancer Center, National Cancer Institute, Nationwide Children's Hospital, Stanford University, University of Alabama, University of Michigan, … Project Team: National Institutes of Health (2017). Integrated genomic characterization of oesophageal carcinoma. Nature, 541(7636), 169–175. https://doi.org/10.1038/nature20805
Song, Y. et al., (2014). Identification of genomic alterations in oesophageal squamous cell cancer. Nature, 509(7498), 91–95. https://doi.org/10.1038/nature13176
Janjigian, Y. Y. et al., (2020). First-line pembrolizumab and trastuzumab in HER2-positive oesophageal, gastric, or gastro-oesophageal junction cancer: an open-label, single-arm, phase 2 trial. The Lancet. Oncology, 21(6), 821–831. https://doi.org/10.1016/S1470-2045(20)30169-8
Lin, D. C. et al., (2014). Genomic and molecular characterization of esophageal squamous cell carcinoma. Nature genetics, 46(5), 467–473. https://doi.org/10.1038/ng.2935

ACKNOWLEDGEMENTS
R01 CA255242-01 NCI (Wei/Paragh;Co-PI) & Roswell Park Alliance Foundation

Sydney Grant